

This is just one of the versions of the exam papers. Your exam question set might have different questions from this one.
This sample solution is for personal use, so please do not share this with others.

Questions 1 to 3 are about the weakest preconditions for the following statement S and postcondition q .

$$S \equiv x := x + 1; y := y \div (x + 1), \quad q \equiv y > 10$$

1. Which one of the following is semantically equivalent to $D(S)$? (4 pts)

- a. $x \neq 0$
- b. $x \neq -1$
- c. $x \neq -2$
- d. $y \neq 10$

2. Calculate $wlp(S, q)$, do not logically simplify your solution. (9 pts)

$$\begin{aligned} wlp(S, q) &\equiv wlp(x := x + 1, y \div (x + 1) > 10) \\ &\equiv y \div ((x + 1) + 1) > 10 \end{aligned}$$

3. Which one of the following is true about the $wlp(S, q)$ and $wp(S, q)$? (4 pts)

- a. $wlp(S, q)$ and $wp(S, q)$ are syntactically equal.
- b. $wlp(S, q)$ and $wp(S, q)$ are semantically equal.
- c. $wlp(S, q)$ is strictly weaker than $wp(S, q)$.
- d. $wlp(S, q)$ is strictly stronger than $wp(S, q)$.

Questions 4 to 8 are about the properties of weakest preconditions and strongest postconditions.

4. Let S be a deterministic program and $M(S, \sigma) \models x > 1$, then which one of the following must be true? (4 pts)

- a. $\sigma \models wp(S, x > 1)$
- b. $\sigma \models wp(S, x \leq 1)$
- c. $\sigma \models wlp(S, x > 1)$
- d. $\sigma \models wlp(S, x \leq 1)$

5. Let S be a program such that $M(S, \sigma) = \{\perp_d\}$, then which one of the following must be true? (4 pts)

- a. $\sigma \models wlp(S, x > 1)$
- b. $\sigma \models wp(S, x \leq 1)$
- c. $\sigma \models wp(S, T)$
- d. $\sigma \models wp(S, F)$

6. Let σ be a proper state for S and $\sigma \models x > 1$, then which one of the following must be true? (4 pts)

- a. $M(S, \sigma) \models sp(x \leq 1, S)$
- b. $M(S, \sigma) \models \neg sp(x \leq 1, S)$
- c. $M(S, \sigma) \models sp(x > 1, S)$
- d. $M(S, \sigma) \models \neg sp(x > 1, S)$

7. Let $\sigma \models p$ and $M(S, \sigma) = \{x = 1\}$, then which one of the following might NOT be true? (4 pts)

- a. $\{x = 1\} \models sp(p, S)$
- b. $\sigma \models sp(p, S)$
- c. $\{x = 1\} \models \{p\} S \{sp(p, S)\}$
- d. $\sigma \models \{p\} S \{sp(p, S)\}$

8. Let $sp(p, S) = x \leq 1$, then which one of the following must be true? (4 pts)
- $wlp(S, x \leq 1) \Rightarrow p$
 - $p \Rightarrow wlp(S, x \leq 1)$
 - $wlp(S, p) \Rightarrow x \leq 1$
 - $x \leq 1 \Rightarrow wlp(S, p)$

Questions 9 and 10 are about the Forward Assignment Axiom.

9. Which one of the following is syntactically equivalent to $sp(p, x := x + 1; x := x + 1)$? (4 pts)
- $p[x + 2/x]$
 - $p[x + 1/x][x + 1/x]$
 - $sp(p, x := x + 2)$
 - $sp(sp(p, x := x + 1), x := x + 1)$

10. Let $p \equiv x > y$, calculate $sp(p, x := x + 1; x := x + 1)$, do not logically simplify your solution. (9 pts)

$$\begin{aligned} sp(p, x := x + 1; x := x + 1) &\equiv sp(x_0 > y \wedge x = x_0 + 1, x := x + 1) \\ &\equiv x_0 > y \wedge x_1 = x_0 + 1 \wedge x = x_1 + 1 \end{aligned}$$

Questions 11 to 14 are about the following minimal proof outline under partial correctness.

$\{T\} \text{ if } b[i] \geq 1 \text{ then } x := \text{sqrt}(b[i]) \text{ fi } \{b[i] \geq 1 \rightarrow x = \text{sqrt}(b[i])\}$

11. Create a **formal proof under partial correctness** for the above minimal proof outline. (11 pts)

1 $\{b[i] \geq 1 \rightarrow \text{sqrt}(b[i]) = \text{sqrt}(b[i])\} x := \text{sqrt}(b[i]) \{b[i] \geq 1 \rightarrow x = \text{sqrt}(b[i])\}$ backward assignment

2 $T \wedge b[i] \geq 1 \Rightarrow b[i] \geq 1 \rightarrow \text{sqrt}(b[i]) = \text{sqrt}(b[i])$ predicate logic

3 $\{T \wedge b[i] \geq 1\} x := \text{sqrt}(b[i]) \{b[i] \geq 1 \rightarrow x = \text{sqrt}(b[i])\}$ strengthen precondition 2,1

4 $\{b[i] \geq 1 \rightarrow x = \text{sqrt}(b[i])\} \text{ skip } \{b[i] \geq 1 \rightarrow x = \text{sqrt}(b[i])\}$ skip

5 $T \wedge b[i] < 1 \Rightarrow b[i] \geq 1 \rightarrow x = \text{sqrt}(b[i])$ predicate logic

6 $\{T \wedge b[i] < 1\} \text{ skip } \{b[i] \geq 1 \rightarrow x = \text{sqrt}(b[i])\}$ strengthen precondition 5,4

7 $\{T\} \text{ if } b[i] \geq 1 \text{ then } x := \text{sqrt}(b[i]) \text{ else skip fi } \{b[i] \geq 1 \rightarrow x = \text{sqrt}(b[i])\}$ if – else 3,6

12. Which one of the following is true about the above formal proof? (4 pts)

- It proves the else branch using the Forward Assignment Axiom.
- It proves the else branch using the Backward Assignment Axiom.
- It proves the conditional statement using the Conditional Rule 1.
- It proves the conditional statement using the Conditional Rule 2.

13. Calculate the domain predicate for the postcondition $b[i] \geq 1 \rightarrow x = \text{sqrt}(b[i])$ in the above proof under partial correctness. (5 pts)

$$\begin{aligned} D(b[i] \geq 1 \rightarrow x = \text{sqrt}(b[i])) &\Leftrightarrow D(b[i] \geq 1) \wedge D(x = \text{sqrt}(b[i])) \\ &\Leftrightarrow 0 \leq i < \text{size}(b) \wedge b[i] \geq 0 \end{aligned}$$

14. Denote the domain predicate you calculated in the above question as d . Create a **full proof outline** for this program **under total correctness** starting with the postcondition. Use **the Backward Assignment Axiom** to prove all assignment statements and use **the Conditional Rule 2** to prove the conditional statement. You don't have to explain the logic implications used in your proof outline. You can use logic variables d, p_1, p_2 and q that are defined in the proof outline directly. (6 pts)

```

{  $0 \leq i < \text{size}(b) \wedge (b[i] \geq 1 \rightarrow p_1) \wedge (b[i] < 1 \rightarrow p_2)$  }
if  $b[i] \geq 1$  then
    {  $p_1 \equiv d \wedge (b[i] \geq 1 \rightarrow \text{sqrt}(b[i]) = \text{sqrt}(b[i]))$  }  $x := \text{sqrt}(b[i])$ 
    {  $q$  }
else
    {  $p_2 \equiv q$  } skip {  $q$  }
fi
{  $q \equiv d \wedge (b[i] \geq 1 \rightarrow x = \text{sqrt}(b[i]))$  }

```

In questions 15 to 17, we are creating a program that calculates the square root of a non-negative integer n without using $\text{sqrt}(n)$. We will use $q \equiv x^2 \leq n < (x + 1)^2$ as the postcondition for this program, where x is the square root of n .

15. Which one of the following can be a precondition for this program? (4 pts)

- a. $n \neq 0$
- b. $n \geq 0$**
- c. $n < 0$
- d. T

16. We will use a loop in this program. Using the technique “replacing a constant in the postcondition by a variable”, create four possible candidates for the loop invariant p and their corresponding loop condition B . You don't need to figure out the range of the introduced variable. (8 pts)

```

 $p_1 \equiv x^2 k \leq n < (x + 1)^2$  and  $B_1 \equiv k \neq 2$ 
 $p_2 \equiv x^2 \leq k < (x + 1)^2$  and  $B_2 \equiv k \neq n$ 
 $p_3 \equiv x^2 \leq n < (x + k)^2$  and  $B_3 \equiv k \neq 1$ 
 $p_4 \equiv x^2 \leq n < (x + 1)^k$  and  $B_4 \equiv k \neq 2$ 

```

17. After considering these candidates created in Question 16, we decided to use the loop invariant created by “replacing constant 1 in the postcondition by a variable k , where k has range $1 \leq k \leq n + 1$ ”. We will “binary search” for x in this program: we start with $x = 0$ and $k = n + 1$; and in the loop, we will shrink the search range by either increasing x or decreasing k until we find some x that satisfies the postcondition.

Here is a **full proof outline under total correctness** for this program. We use **the Forward Assignment Axiom to prove the loop initialization**; and we use **the Condition Rule 1 and the Backward Assignment Axiom to prove the loop body**. Finish this full proof outline under total correctness for this program by calculating the missing conditions and expressions. You don’t need to explain the logic implications used in the proof outline. You can use logic variables p, t and p_1 that are defined in the proof outline directly. (22 pts)

```

{  $n \geq 0$  }  $x := 0$ ; {  $n \geq 0 \wedge x = 0$  }

 $k := n + 1$ ; {  $n \geq 0 \wedge x = 0 \wedge k = n + 1$  }

{inv  $p \equiv x^2 \leq n < (x + k)^2 \wedge 1 \leq k \leq n + 1$ } {bd  $t \equiv k$ }

while  $k \neq 1$  do {  $p_1 \equiv p \wedge k \neq 1 \wedge t = t_0$  }

    if  $(x + k \div 2)^2 > n$  then {  $p_1 \wedge (x + k \div 2)^2 > n$  }

        {  $x^2 \leq n < (x + k \div 2)^2 \wedge 1 \leq k \div 2 \leq n + 1 \wedge k \div 2 < t_0$  }

         $k := k \div 2$  {  $p \wedge t < t_0$  }

    else {  $p_1 \wedge (x + k \div 2)^2 \leq n$  }

        {  $(x + k \div 2)^2 \leq n < (x + k)^2 \wedge 1 \leq k - k \div 2 \leq n + 1 \wedge k - k \div 2 < t_0$  }  $x := x + k \div$ 
2;

        {  $x^2 \leq n < (x + k - k \div 2)^2 \wedge 1 \leq k - k \div 2 \leq n + 1 \wedge k - k \div 2 < t_0$  }

         $k := k - k \div 2$  {  $p \wedge t < t_0$  }

    {  $p \wedge t < t_0$  } fi

od {  $x^2 \leq n < (x + k)^2 \wedge 1 \leq k \leq n + 1 \wedge k = 1$  }
{  $x^2 \leq n < (x + 1)^2$  }

```